BITS PILANI, DUBAI CAMPUS

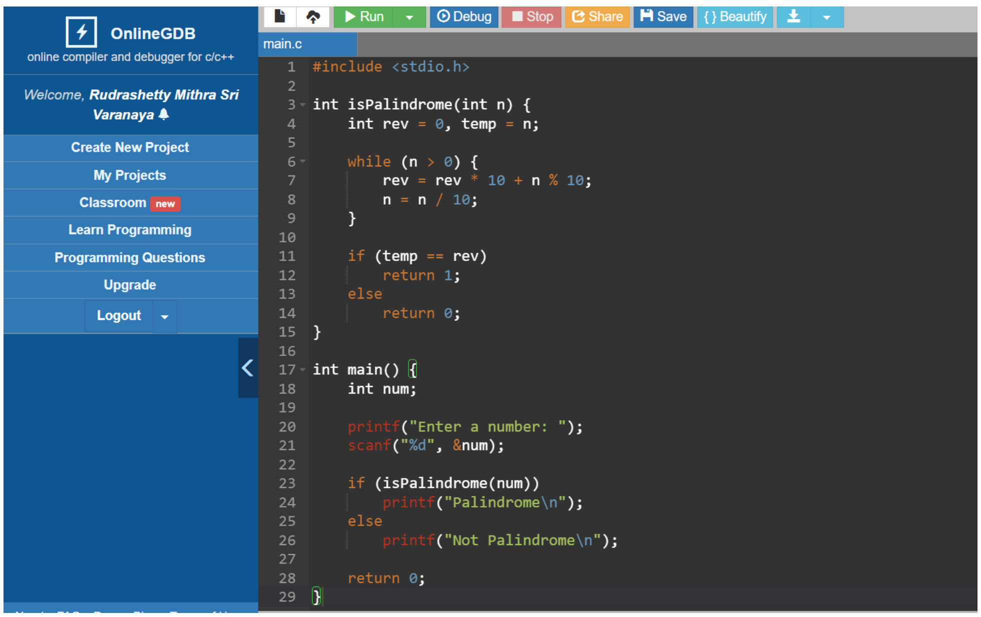
DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI

COURSE	: COMPUTER PROGRAMMING (CS F111)
COMPONENT	: Practical Week 8: User defined functions

1) Write a C program using a function to check whether a given number is palindrome or not.

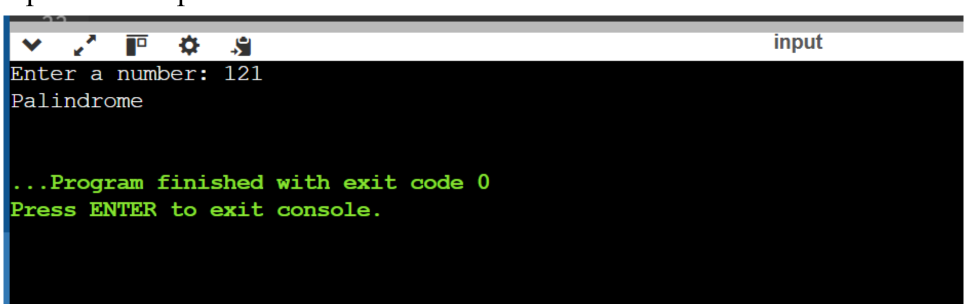
Answer:

Code:




```c
#include <stdio.h>

int isPalindrome(int n) {
    int rev = 0, temp = n;

    while (n > 0) {
        rev = rev * 10 + n % 10;
        n = n / 10;
    }

    if (temp == rev)
        return 1;
    else
        return 0;
}

int main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    if (isPalindrome(num))
        printf("Palindrome\n");
    else
        printf("Not Palindrome\n");

    return 0;
}
```

Input and Output:




```
Enter a number: 121
Palindrome


...Program finished with exit code 0
Press ENTER to exit console.
```

2)Write a C program using a function to reverse a given number.

Answer:

Code:

```c
#include <stdio.h>

int reverse(int n) {
    int rev = 0;

    while (n > 0) {
        rev = rev * 10 + n % 10;
        n = n / 10;
    }

    return rev;
}

int main() {
    int num;

    printf("Enter a number: ");
    scanf("%d", &num);

    printf("Reversed number = %d\n", reverse(num));

    return 0;
}
```

Input and Output:

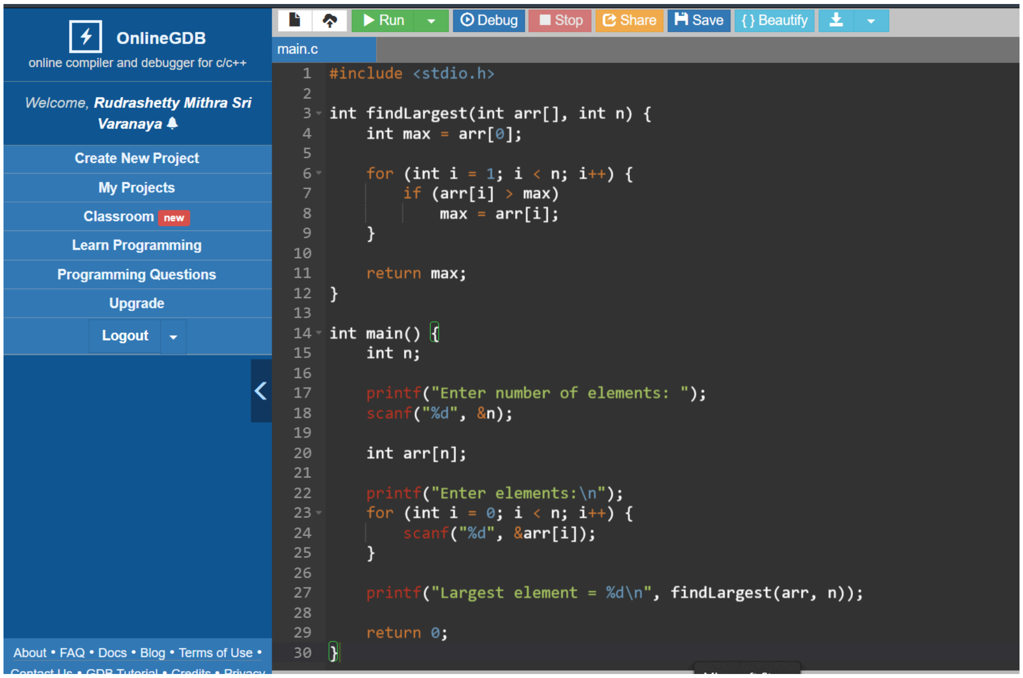
```
Enter a number: 1234
Reversed number = 4321


...Program finished with exit code 0
Press ENTER to exit console.
```

---

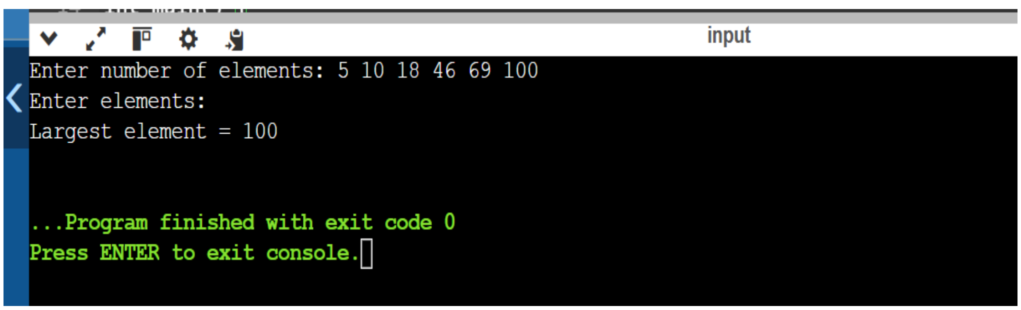
3)Write a C program to pass an array to a function and find the largest element in the array.

Answer:

Code: Scroll Down,



Input and Output:

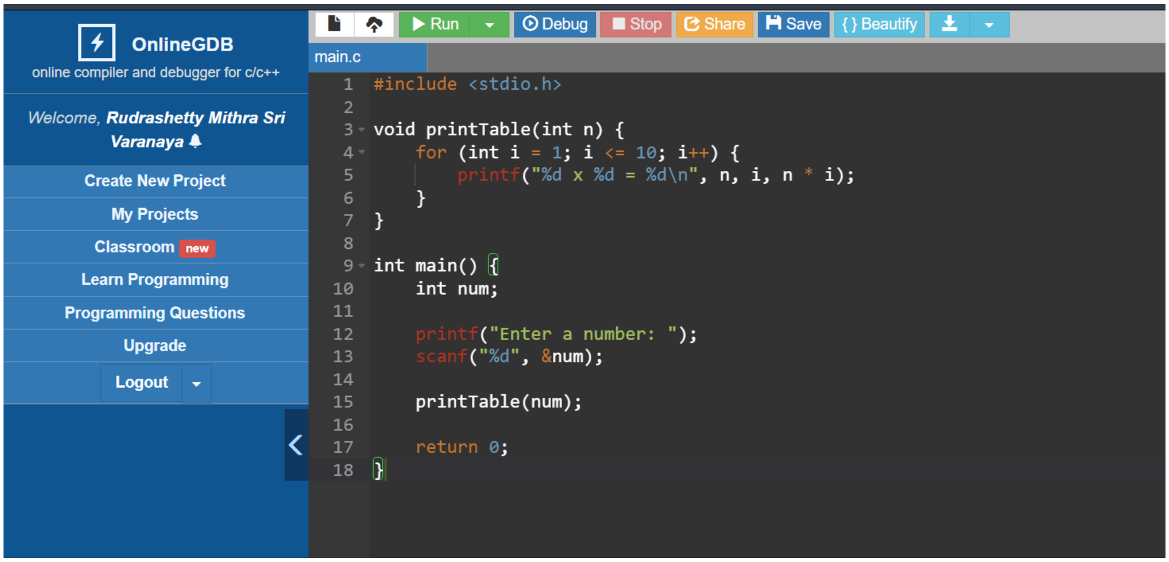


---

4)Write a C program using a function to print the multiplication table of a given number.

Answer:

Code: Scroll down,



Input and Output: